CSE 270 Sports Analytics

Measuring individual performance in team sports

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Plus/Minus

Plus/Minus

The concept

Plus-minus measures the point differential for the team when the player is on the court,

Point differential is the difference in the points scored for and the points scored against the team.

The NHL's Montreal Canadiens were the first team to track the plus - minus of its players, starting sometime in the 1950s. Other teams followed in the early 1960s, and the NHL started officially compiling the statistic for the 1967 - 68 season.

Plus/minus for NBA

```
colnames(nba12)
```

```
##  [1] "GameID"           "Date"             "AwayT"            "HomeT"
##  [5] "StartTime"        "EndTime"          "ElapsedTime"      "ElapsedSecs"
##  [9] "HomePlayer1ID"    "HomePlayer2ID"    "HomePlayer3ID"    "HomePlayer4ID"
## [13] "HomePlayer5ID"    "AwayPlayer1ID"    "AwayPlayer2ID"    "AwayPlayer3ID"
## [17] "AwayPlayer4ID"    "AwayPlayer5ID"    "HomePlayer1Name"  "HomePlayer2Name"
## [21] "HomePlayer3Name"  "HomePlayer4Name"  "HomePlayer5Name"  "AwayPlayer1Name"
## [25] "AwayPlayer2Name"  "AwayPlayer3Name"  "AwayPlayer4Name"  "AwayPlayer5Name"
## [29] "StartScoreHome"   "StartScoreAway"   "EndScoreHome"     "EndScoreAway"
## [33] "PointsScoredHome" "PointsScoredAway" "PossessionsHome"  "PossessionsAway"
```

Plus/minus for NBA

Subseting the data frame for Oklahoma City Thunders

```
oks_g <- grepl("OKC", nba12$GameID, fixed = TRUE)
okc<-nba12[oks_g == T,]
```

```
head(okc[,19:23], n=5)
```

```
##     HomePlayer1Name HomePlayer2Name HomePlayer3Name HomePlayer4Name
## 144       Westbrook       Sefolosha          Durant           Ibaka
## 145       Westbrook       Sefolosha          Durant           Ibaka
## 146       Westbrook          Harden          Durant           Ibaka
## 147       Westbrook          Harden          Durant           Ibaka
## 148       Westbrook          Harden          Durant           Ibaka
##     HomePlayer5Name
## 144         Perkins
## 145         Perkins
## 146         Perkins
## 147        Collison
## 148        Collison
```

Plus/minus for NBA

Create the line-ups variable

```
okc$line_h <-
  paste(okc$HomePlayer1Name, okc$HomePlayer2Name,     okc$HomePlayer3Name, okc$HomeP
        okc$HomePlayer5Name, sep = ",")

okc$line_a <-
  paste(okc$AwayPlayer1Name, okc$AwayPlayer2Name, okc$AwayPlayer3Name,okc$AwayPlayer
        okc$AwayPlayer5Name,   sep = ",")

okc$line_h[1:10]
```

```
##  [1] "Westbrook,Sefolosha,Durant,Ibaka,Perkins"
##  [2] "Westbrook,Sefolosha,Durant,Ibaka,Perkins"
##  [3] "Westbrook,Harden,Durant,Ibaka,Perkins"
##  [4] "Westbrook,Harden,Durant,Ibaka,Collison"
##  [5] "Westbrook,Harden,Durant,Ibaka,Collison"
##  [6] "Westbrook,Harden,Durant,Ibaka,Collison"
##  [7] "Maynor,Harden,Durant,Collison,Mohammed"
##  [8] "Maynor,Harden,Durant,Collison,Mohammed"
##  [9] "Maynor,Cook,Harden,Collison,Mohammed"
## [10] "Maynor,Cook,Harden,Collison,Mohammed"
```

Plus/minus for NBA

Calculating plus/minus variable for time stunts

Away team

```
# Home team plus minus
okc$Home_PM <- okc$PointsScoredHome - okc$PointsScoredAway
# Away team plus minus
okc$Away_PM <- -okc$Home_PM
```

Check if Durant is playing

```
# is Durant playing at home?
okc$D_H <- grepl("Durant", okc$line_h, fixed = TRUE)
# is Durant playing away?
okc$D_A <- grepl("Durant", okc$line_a, fixed = TRUE)
```

Plus/minus for NBA

Get the plus minus for Durant for the whole season

```
sum(okc$Home_PM[okc$D_H == T]) + sum(okc$Away_PM[okc$D_A == T])
```

```
## [1] 324
```

This will be the plus/minus for the whole season

Plus/minus for NBA

For each game

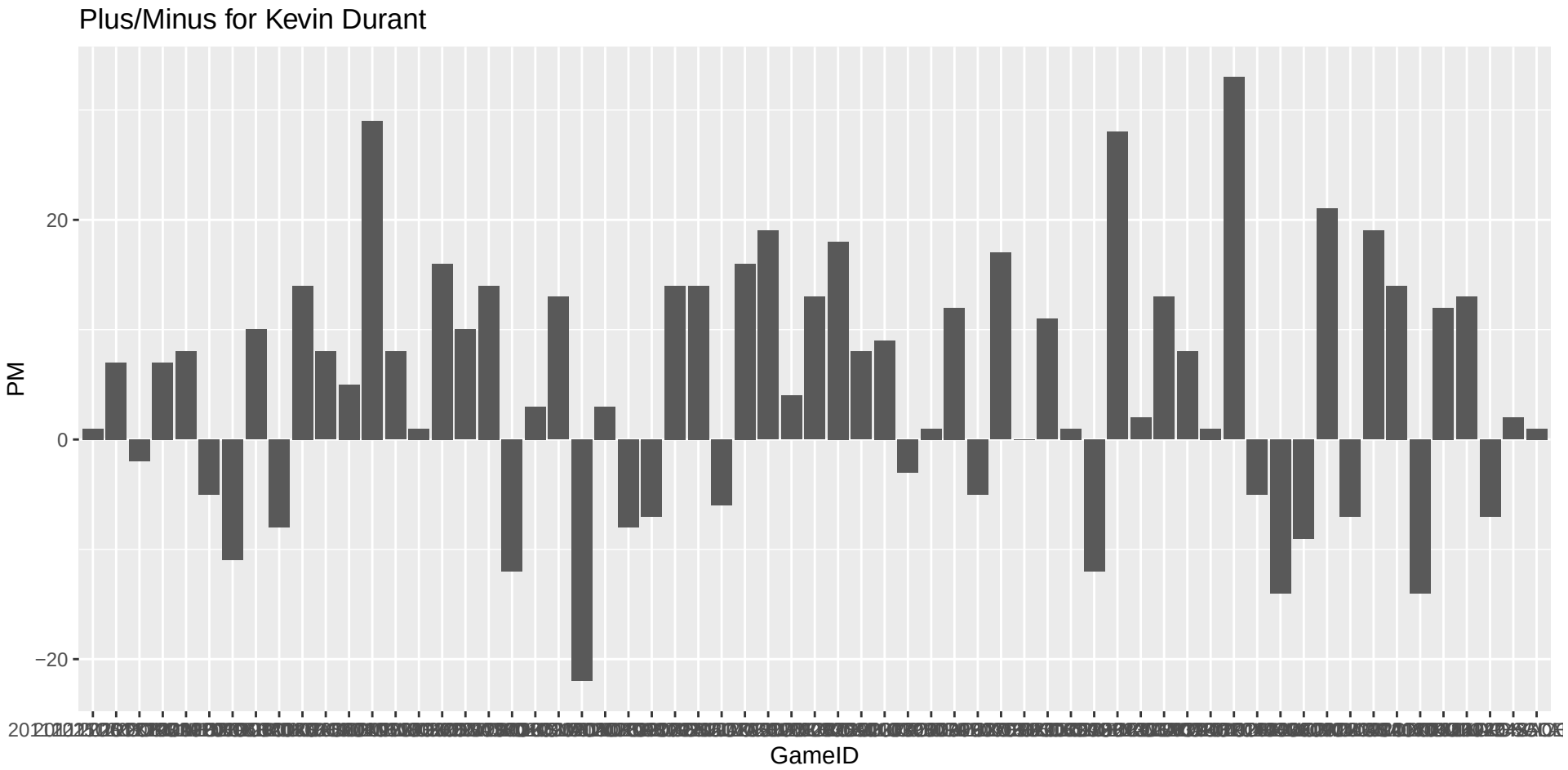
```
durant_pm <- okc %>%
  mutate(PM=ifelse(D_H==T,
                   Home_PM,
                   ifelse(D_A==T, Away_PM,0))) %>%
  group_by(GameID) %>%
  summarise(PM=sum(PM))
```

```
head(durant_pm, n=5)
```

```
## # A tibble: 5 x 2
##   GameID              PM
##   <fct>            <dbl>
## 1 20111225ORLOKC       1
## 2 20111226OKCMIN       7
## 3 20111228OKCMEM      -2
## 4 20111229DALOKC       7
## 5 20111231PHXOKC       8
```

Plus/minus for NBA

```
ggplot(durant_pm, aes(x=GameID, y=PM))+
  geom_bar(stat="identity")+
  ggtitle("Plus/Minus for Kevin Durant")
```



On court and off court +/- and Roland's rating

The on court +/- number represents the team's net points with the player on the floor per 48 minutes, while the off court number is the team's net with the player off the floor per 48 minutes.

The Roland Rating is the difference between the two, with a positive number indicating the team has played better with the player than without.

Net Plus/minus for NBA

Create a variable that shows OKC plus/minus, depending on whether it is playing at home or away

```
okc <- okc %>%
# makes PM for OKC
mutate(OKC_PM = ifelse(AwayT == "OKC", Away_PM, ifelse(HomeT == "OKC", Home_PM, 0)),
       D_PM = ifelse(D_H == T | D_A == T, T, F)
       )
```

Net Plus/minus for NBA

For each game, D_PM has a value of FALSE if Durant has not played, and TRUE if He has. PM is showing the plus/minus for that game

```
durant_pm <- okc %>%
group_by(GameID, D_PM) %>%
summarise(PM = sum(OKC_PM))
head(durant_pm)
```

```
## # A tibble: 6 x 3
## # Groups:   GameID [3]
##   GameID         D_PM     PM
##   <fct>          <lgl> <dbl>
## 1 20111225ORLOKC FALSE     7
## 2 20111225ORLOKC TRUE      1
## 3 20111226OKCMIN FALSE    -3
## 4 20111226OKCMIN TRUE      7
## 5 20111228OKCMEM FALSE     5
## 6 20111228OKCMEM TRUE     -2
```

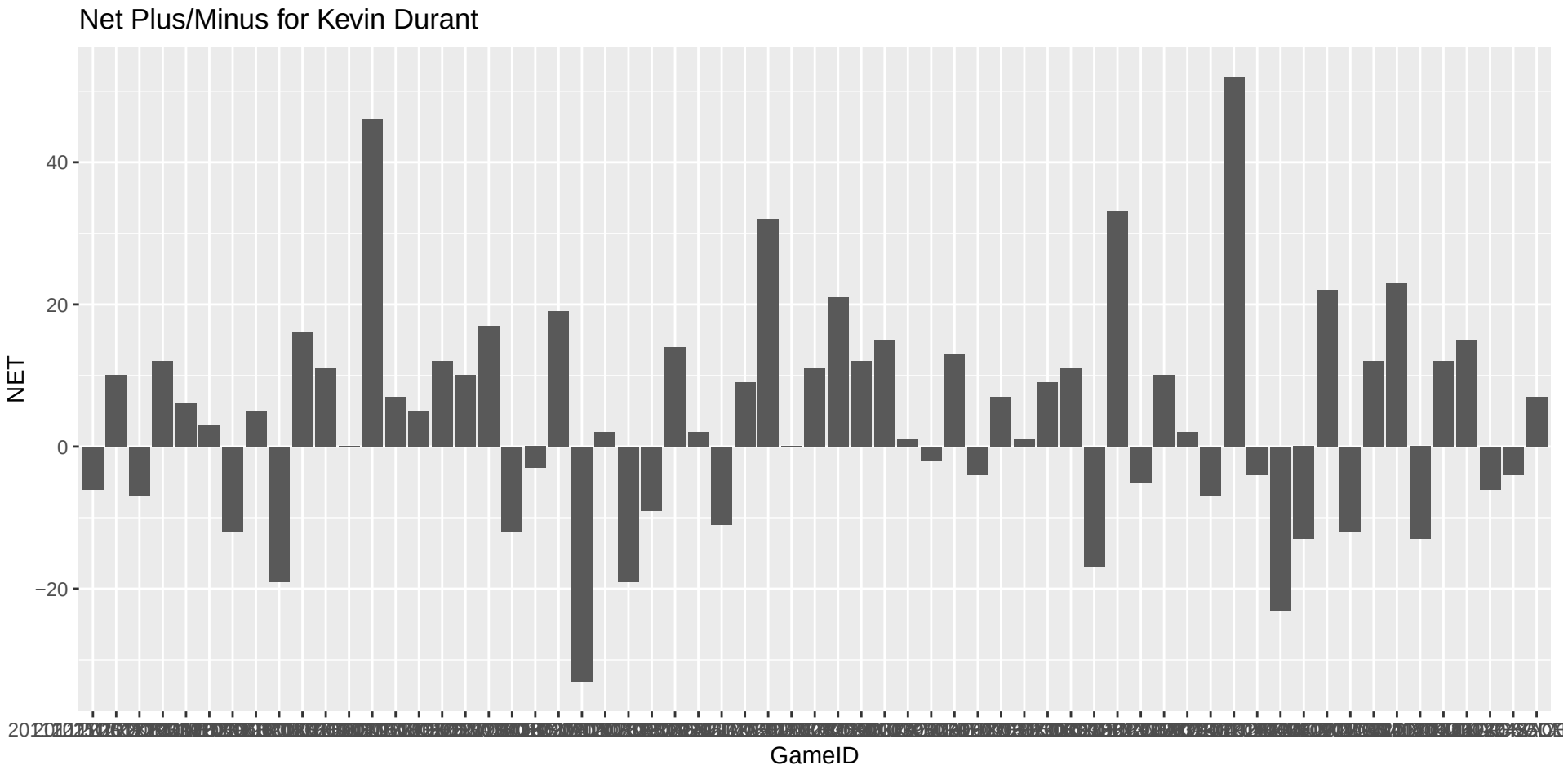
Net Plus/minus for NBA

```
durant_pm <- okc %>%
group_by(GameID, D_PM) %>%
summarise(PM = sum(OKC_PM)) %>%
group_by(GameID) %>%
mutate(NET = PM - lag(PM, 1)) %>%
filter(NET != "NA") %>%
select(GameID, NET)
head(durant_pm)
```

```
## # A tibble: 6 x 2
## # Groups:   GameID [6]
##   GameID           NET
##   <fct>          <dbl>
## 1 20111225ORLOKC    -6
## 2 20111226OKCMIN    10
## 3 20111228OKCMEM    -7
## 4 20111229DALOKC    12
## 5 20111231PHXOKC     6
## 6 20120102OKCDAL     3
```

Plus/minus for NBA

```
ggplot(durant_pm, aes(x=GameID, y=NET))+
  geom_bar(stat="identity")+
  ggtitle("Net Plus/Minus for Kevin Durant")
```



Adjusted plus/minus

Adjusted plus/minus

Adjusted plus/minus

The plus/minus rating is adjusted by the players who are on the court

$$MARGIN = \beta_0 + \beta_1 X_1 + \beta_2 X_@ + ...\beta_k X_k + \epsilon$$

where

$X_k = 1$, if player k is playing at home,
$X_k = -1$, if player k is playing away,
$X_k = 0$, if player k is not playing

Adjusted plus/minus

According to Dan Rosenbaum's approach, β coefficients will show the adjusted plus/minus of the players, or

β_k measures the difference between player k and the reference players, holding the other k players constant

Data preparation/ this will take at least 30-40 minutes on the average machine

Data prep for adjusted plus/minus

```
# Get home and away teams
nba3 <- nba12[, c("HomePlayer1ID", "HomePlayer2ID",
                  "HomePlayer3ID", "HomePlayer4ID", "HomePlayer5ID")]

nba4 <- nba12[, c("AwayPlayer1ID", "AwayPlayer2ID",
                  "AwayPlayer3ID", "AwayPlayer4ID","AwayPlayer5ID")]

# MAke ID as character
nba3 <- as.data.frame(sapply(nba3, as.character))
nba4 <- as.data.frame(sapply(nba4, as.character))

#create dummy variables
nba3 <- as.data.frame(model.matrix(~. - 1, nba3))
nba4 <- as.data.frame(model.matrix(~. - 1, nba4))
# for away team
nba4[nba4==1]<- -1
# change colnames to be more readable
colnames(nba3) <- gsub("HomePlayer[123456789]ID", "", colnames(nba3))
colnames(nba4) <- gsub("AwayPlayer[123456789]ID", "", colnames(nba4))

# nba3 remove double entries
nba3 <- t(nba3)     # transpose the data.frame
nba3 <- by(nba3, INDICES=row.names(nba3), FUN=colSums) # collapse the rows with the same name
nba3 <- as.data.frame(do.call(cbind,aggr))
nba4 <- t(nba4)     # transpose the data.frame
nba4 <- by(nba4, INDICES=row.names(nba4), FUN=colSums) # collapse the rows with the same name
nba4 <- as.data.frame(do.call(cbind,nba4))

#combine home and away players and double columns
nba5<-data.frame(nba3,nba4)
nba5 <- t(nba5)     # transpose the data.frame
nba5 <- by(nba5, INDICES=row.names(nba5), FUN=colSums) # collapse the rows with the same name
nba5 <- as.data.frame(do.call(cbind,nba5))
```

Adjusted plus/minus

```
load("./data/nba5.rda")
```

```
dim(nba5)
```

```
## [1] 25300   474
```

Adjusted plus/minus

Coefficients are named by player ID, we need to match with player names

Add plus/minus as dependent variable

```
nba5<-data.frame(PM=nba12$PointsScoredHome-nba12$PointsScoredAway, nba5)
```

Run the regression

```
model<-lm(PM~., data=nba5)
coeff<-sort(coefficients(model), decreasing=T)
coeff[1:10]
```

```
##     X1027      X453      X936      X678      X900      X158      X904      X781
## 2.1176032 1.8681832 1.5567575 1.0007997 0.9188331 0.8786333 0.8486712 0.8390129
##      X919      X964
## 0.8211618 0.8188564
```

Remove X from the begining of the coeff name and match with player info

```
name1<-names(coeff)
name1<-gsub("X", "", name1)
name1<-as.data.frame(name1)
```

Adjusted plus/minus

Corresponding dataframe with player level information

```
load("./data/nba12p.rda")
head(nba12p[,1:3],5)
```

```
##   PlayerID   PlayerTrueName PlayerTeams
## 1        3   Dampier, Erick         ATL
## 2       17   Diop, DeSagana         CHA
## 3       35    Boykins, Earl         HOU
## 4       36  Najera, Eduardo         CHA
## 5       42 Elson, Francisco         PHI
```

Adjusted plus/minus

```
nba12p$PlayerID<-as.character(nba12p$PlayerID)
name2<-left_join(name1, nba12p[,1:2], by=c("name1"="PlayerID"))
name2[1:5,]
```

```
##   name1    PlayerTrueName
## 1  1027     Benson, Keith
## 2   453        Diogu, Ike
## 3   936   N'Diaye, Hamady
## 4   678    Williams, Sean
## 5   900 Whiteside, Hassan
```

rename coefficients by names

```
names(coeff)<-name2$PlayerTrueName
```

Adjusted plus/minus

Do you know this players?

Lets look at top ten players

```
coeff[1:10]
```

```
##     Benson, Keith         Diogu, Ike   N'Diaye, Hamady    Williams, Sean
##         2.1176032          1.8681832          1.5567575          1.0007997
## Whiteside, Hassan    Nelson, Jameer     Bledsoe, Eric      Lopez, Brook
##         0.9188331          0.8786333          0.8486712          0.8390129
##     Orton, Daniel   Macklin, Vernon
##         0.8211618          0.8188564
```

Benson, Keith - https://goo.gl/YxMsJ9

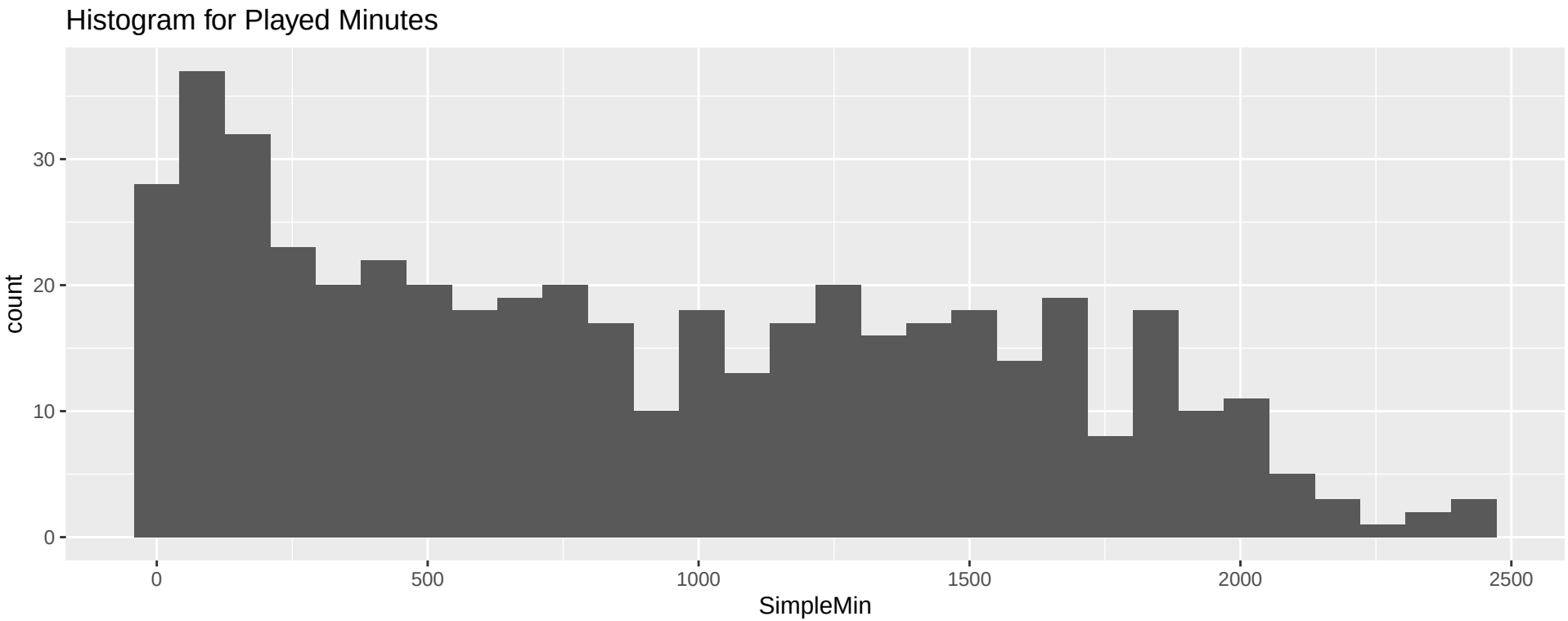
Adjusted plus/minus

Leave only players that has played more than 250 minutes (By Rosenbaum)

```
summary(nba12p$SimpleMin)
```

```
##     Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      0.0   292.1   812.5   892.0  1416.6  2430.9
```

```
ggplot(data=nba12p, aes(SimpleMin))+geom_histogram()+
  ggtitle("Histogram for Played Minutes")
```



Adjusted plus/minus

Getting players with more than 250m of play

```
more250<-nba12p$PlayerID[nba12p$SimpleMin>250]
length(more250)
```

```
## [1] 372
```

```
more250[1:10]
```

```
##  [1] "17"  "51"  "109" "132" "144" "211" "459" "484" "514" "550"
```

```
# Subset the dataframe by more250
colnames(nba5) <- gsub("X", "", colnames(nba5))
nba6<-nba5[,colnames(nba5) %in% more250]
nba6<-data.frame(PM=nba5$PM, nba6)
# Make the model
model2<-lm(PM~., data=nba6)
# get the coefficients
coeff2<-sort(coefficients(model2), decreasing=T)
coeff2[1:10]
```

```
##      X158      X113      X807       X70      X144      X904      X821      X791
## 0.8779187 0.7791911 0.7605357 0.6760659 0.6705972 0.6580038 0.6435656 0.6088434
##      X938        X5
## 0.6001202 0.5689375
```

Adjusted plus/minus

Get the adjust plus/minus from coefficients

```
name1<-names(coeff2)
name1<-gsub("X", "", name1)
name1<-as.data.frame(name1)
name2<-left_join(name1, nba12p[,1:2], by=c("name1"="PlayerID"))
names(coeff2)<-name2$PlayerTrueName
```

Adjusted plus/minus

```
coeff2[1:10]
```

```
## Nelson, Jameer   Duhon, Chris Hudson, Lester    Paul, Chris   Battie, Tony
##      0.8779187      0.7791911      0.7605357      0.6760659      0.6705972
##  Bledsoe, Eric Curry, Stephen   Lopez, Robin     Wall, John Nowitzki, Dirk
##      0.6580038      0.6435656      0.6088434      0.6001202      0.5689375
```

Nelson Jameer https://goo.gl/eNj7sq

Adjusted plus/minus

Where is Durant ?

```
name2[name2$PlayerTrueName=="Durant, Kevin",]
```

```
##     name1 PlayerTrueName
## NA   <NA>           <NA>
## 225   714  Durant, Kevin
```

Where is Lebron ?

```
name2[name2$PlayerTrueName=="James, LeBron",]
```

```
##    name1 PlayerTrueName
## 15   260  James, LeBron
## NA  <NA>           <NA>
```

Adjusted plus/minus

- Modifications
  - Take only players that played more then 250 minutes during one season
  - As a depended variable, take margin of Plus/minus per 100 possessions

$$\text{Margin} = \frac{\text{Home Points}}{\text{Home Possession}} - \frac{\text{Away Points}}{\text{Away Possesssion}}$$

Adjusted Plus/Minus

```
str(nba12)
```

```
## 'data.frame':    25300 obs. of  36 variables:
##  $ GameID          : Factor w/ 883 levels "20111225BOSNYK",..: 1 1 1 1 1 1 1 1 1 1 ...
##  $ Date            : chr  "20111225" "20111225" "20111225" "20111225" ...
##  $ AwayT           : chr  "BOS" "BOS" "BOS" "BOS" ...
##  $ HomeT           : chr  "NYK" "NYK" "NYK" "NYK" ...
##  $ StartTime       : chr  "0:48:00" "0:42:59" "0:42:17" "0:41:42" ...
##  $ EndTime         : chr  "0:42:59" "0:42:17" "0:41:42" "0:40:35" ...
##  $ ElapsedTime     : chr  "0:05:01" "0:00:42" "0:00:35" "0:01:07" ...
##  $ ElapsedSecs     : int  301 42 35 67 58 35 146 30 6 165 ...
##  $ HomePlayer1ID   : int  843 843 843 990 990 990 990 990 990 843 ...
##  $ HomePlayer2ID   : int  915 915 915 915 752 752 752 25 25 990 ...
##  $ HomePlayer3ID   : int  25 25 25 25 25 25 25 315 315 315 ...
##  $ HomePlayer4ID   : int  537 537 537 315 315 315 315 597 597 537 ...
##  $ HomePlayer5ID   : int  118 118 118 118 118 988 988 988 988 597 ...
##  $ AwayPlayer1ID   : int  553 553 553 553 553 553 876 876 553 553 ...
##  $ AwayPlayer2ID   : int  166 166 166 166 419 419 419 419 876 876 ...
##  $ AwayPlayer3ID   : int  265 14 14 14 14 14 14 14 14 14 ...
##  $ AwayPlayer4ID   : int  285 83 181 181 181 181 181 181 181 181 ...
##  $ AwayPlayer5ID   : int  143 285 83 83 83 83 285 285 285 285 ...
##  $ HomePlayer1Name : chr  "Douglas" "Douglas" "Douglas" "Shumpert" ...
##  $ HomePlayer2Name : chr  "Fields" "Fields" "Fields" "Fields" ...
##  $ HomePlayer3Name : chr  "Anthony" "Anthony" "Anthony" "Anthony" ...
##  $ HomePlayer4Name : chr  "Stoudemire" "Stoudemire" "Stoudemire" "Jeffries" ...
##  $ HomePlayer5Name : chr  "Chandler" "Chandler" "Chandler" "Chandler" ...
##  $ AwayPlayer1Name : chr  "Rondo" "Rondo" "Rondo" "Rondo" ...
##  $ AwayPlayer2Name : chr  "Allen" "Allen" "Allen" "Allen" ...
##  $ AwayPlayer3Name : chr  "Pavlovic" "Daniels" "Daniels" "Daniels" ...
##  $ AwayPlayer4Name : chr  "Garnett" "Bass" "Wilcox" "Wilcox" ...
##  $ AwayPlayer5Name : chr  "O'Neal" "Garnett" "Bass" "Bass" ...
##  $ StartScoreHome  : int  0 14 17 19 19 21 24 32 34 34 ...
##  $ StartScoreAway  : int  0 9 12 12 14 16 16 22 23 23 ...
##  $ EndScoreHome    : int  14 17 19 19 21 24 32 34 34 41 ...
##  $ EndScoreAway    : int  9 12 12 14 16 16 22 23 23 29 ...
##  $ PointsScoredHome: int  14 3 2 0 2 3 8 2 0 7 ...
##  $ PointsScoredAway: int  9 3 0 2 2 0 6 1 0 6 ...
##  $ PossessionsHome : int  9 2 1 1 2 1 6 1 1 5 ...
```

Adjusted Plus/Minus

PM for possessions

```
nba12$PM_Home<-nba12$PointsScoredHome/nba12$PossessionsHome
nba12$PM_Away<-nba12$PointsScoredAway/nba12$PossessionsAway
nba12$Margin<-nba12$PM_Home-nba12$PM_Away
summary(nba12$PM_Home)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
##    0.00    0.50    1.00     Inf    1.62     Inf    2085
```

```
summary(nba12$PM_Away)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
##  0.0000  0.4444  1.0000     Inf  1.5000     Inf    2120
```

```
summary(nba12$Margin)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
##    -Inf  -0.667   0.000     NaN   0.692     Inf    3576
```

We have choice, remove cases with not number values, or impute with the mean

We will remove

```
nba6$PM<-nba12$Margin
nba6$Secs<-nba12$ElapsedSecs/60
nba7<-nba6[is.finite(nba6$PM),]
dim(nba7)
```

```
## [1] 21642   373
```

Adjusted Plus/Minus

The model

```
model3<-lm(PM~.,data=nba7)

coeff3<-sort(coefficients(model3), decreasing=T)
# basic cleaning
name1<-names(coeff3)
name1<-gsub("X", "", name1)
name1<-as.data.frame(name1)
name2<-left_join(name1, nba12p[,1:2], by=c("name1"="PlayerID"))
names(coeff3)<-name2$PlayerTrueName
```

Adjusted Plus/Minus

```
# top 10
coeff3[1:10]
```

```
##     Nash, Steve  Nelson, Jameer    Mahinmi, Ian  Hudson, Lester    Duhon, Chris
##       0.3479366       0.3374820       0.3280165       0.3243823       0.2744698
##   Millsap, Paul   Kapono, Jason Humphries, Kris    Rondo, Rajon  Curry, Stephen
##       0.2454976       0.2445955       0.2423881       0.2405524       0.2374232
```

Steve Nash, Phoenix, https://goo.gl/BczeBv

Adjusted Plus/Minus

Interpretations: If you take 4 average players and Steve Nash than the team will make by 0.35 points/per possession more then another team with 5 average players

```
# top 10
coeff3[1:10]
```

```
##     Nash, Steve  Nelson, Jameer    Mahinmi, Ian  Hudson, Lester    Duhon, Chris
##       0.3479366       0.3374820       0.3280165       0.3243823       0.2744698
##   Millsap, Paul   Kapono, Jason Humphries, Kris    Rondo, Rajon  Curry, Stephen
##       0.2454976       0.2445955       0.2423881       0.2405524       0.2374232
```

Adjusted Plus/Minus

Where is Durant ?

```
name2[name2$PlayerTrueName=="Durant, Kevin",]
```

```
##       name1 PlayerTrueName
## 102     714  Durant, Kevin
## NA     <NA>           <NA>
## NA.1   <NA>           <NA>
```

Where is LeBron ?

```
name2[name2$PlayerTrueName=="James, LeBron",]
```

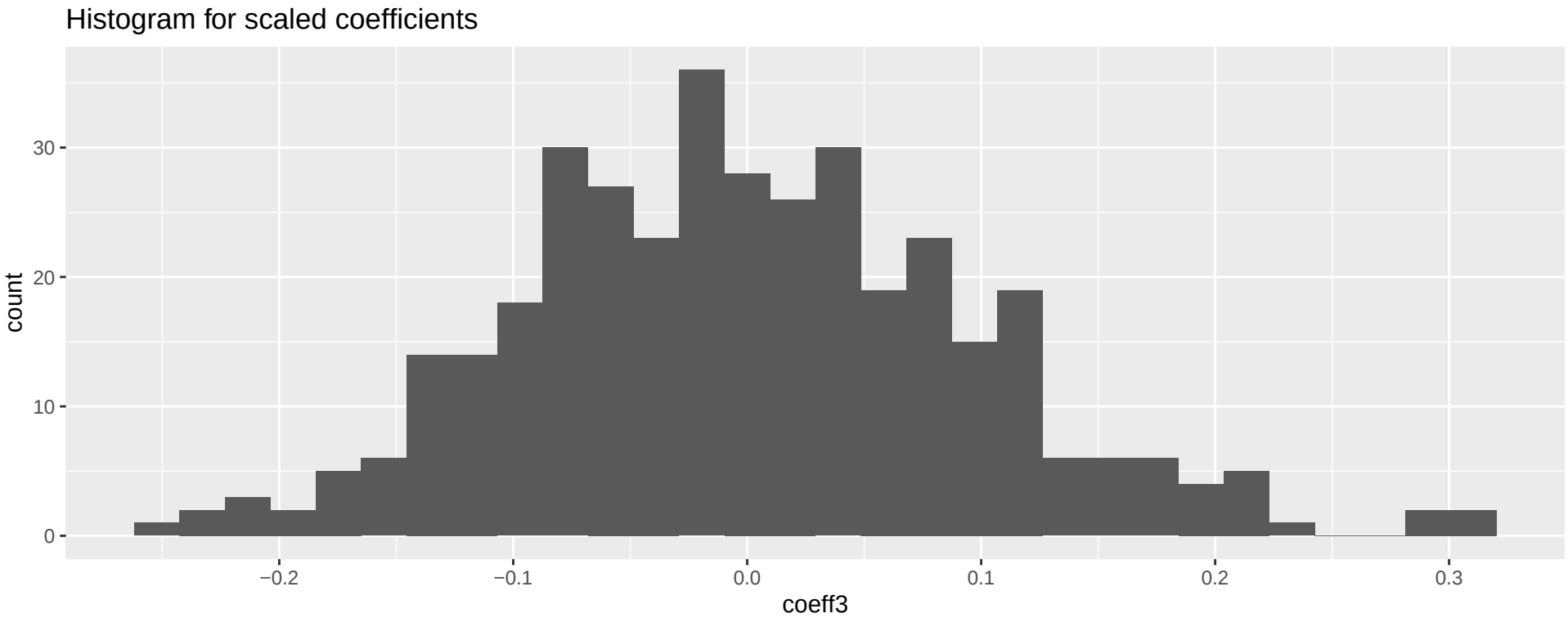
```
##       name1 PlayerTrueName
## 20      260  James, LeBron
## NA     <NA>           <NA>
## NA.1   <NA>           <NA>
```

Adjusted Plus/Minus

We can scale the Plus/Minus values around the mean, such as the positive coefficient will show that the player is doing better than the average and negative will mean that the player is doing worst than the average.

```
coeff3<-coeff3-mean(coeff3)
```

```
ggplot()+aes(coeff3)+geom_histogram()+
  ggtitle("Histogram for scaled coefficients")
```



Other considerations for regression

- Weighted by the duration of the stint, so longer time periods have bigger weights
- Weighting cases by clutch time vs garbage time

Clutch Time: During the 4th quarter or overtime, with less than five minutes remaining, and neither team ahead by more than X points.

Garbage Time: 4th quarter and overtime where either team has a lead of X points

The value of X varies for different analyses, most often 5 and 10.

Weighted regression

```
nba7$Secs<-nba7$Secs/60
model4<-lm(PM~.,data=nba7, weights = Secs)
coeff4<-sort(coefficients(model4), decreasing=T)
# basic cleaning
name1<-names(coeff4)
name1<-gsub("X", "", name1)
name1<-as.data.frame(name1)
name2<-left_join(name1, nba12p[,1:2], by=c("name1"="PlayerID"))
names(coeff4)<-name2$PlayerTrueName
coeff[1:10]
```

```
##     Benson, Keith         Diogu, Ike   N'Diaye, Hamady    Williams, Sean
##         2.1176032          1.8681832          1.5567575          1.0007997
## Whiteside, Hassan    Nelson, Jameer     Bledsoe, Eric      Lopez, Brook
##         0.9188331          0.8786333          0.8486712          0.8390129
##     Orton, Daniel   Macklin, Vernon
##         0.8211618          0.8188564
```

Adjusted Plus/Minus

- The APM is not an indicator on how good or bad the player is.
- The APM shows how helpful where the players for their team success.

APM regressed against other statistics

Dan Rosenbaum suggests to run the following regression to understand the impact of the box score statistics on APM

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_@ + ...\beta_k X_k + \epsilon$$

Where Y is the coefficients of the players from Plus/Minus model,

$X_1, ..X_k$ are box score statistics

APM regressed against other statistics

The results of the regression are the following:

Each coefficient shows the effect of the players box-score statistics on the APM

Basketball Statistic	Coefficient Estimate	Standard Error	Sample Mean	Standard Deviation
Pure adjusted plus/minus rating (dependent variable)	N/A		0.00	4.99
Points per 40 minutes	0.757	0.231	15.55	4.37
Effective field goal attempts per 40 minutes	-1.879	0.402	15.03	3.89
(Effective field goal attempts per 40 minutes)^2	0.026	0.010	241.1	123.3
Three point attempts per 40 minutes	0.323	0.139	2.46	2.21
Free throw attempts per 40 minutes	0.366	0.217	4.02	1.78
Assists per 40 minutes	-0.293	0.277	3.54	2.09
Offensive rebounds per 40 minutes	0.054	0.333	2.00	1.23
Defensive rebounds per 40 minutes	-0.615	0.271	4.99	1.82
Turnovers per 40 minutes	-2.171	0.480	2.36	0.65
Steals per 40 minutes	2.166	0.496	1.31	0.45
Blocks per 40 minutes	1.564	0.327	0.84	0.84
Personal fouls per 40 minutes	0.392	0.281	3.57	1.13
(Points per 40 * Assists per 40 * Rebounds per 40)$^{1/3}$	2.262	0.486	6.71	1.38
Minutes per game	0.108	0.037	27.48	8.44
Intercept	-0.043	3.083	N/A	
R^2	0.4397			
Sample size	420			

Other metrics

- Box-Score PM (Statistical Plus/Minus)- use Box-Score statistics for each stint of time of time for each player as inputs (steals, rebounds, offensive rating, etc)
- ESPN-Real Plus/Minus – is calculated by ESPN again using some sort of regression analysis.
- Offensive Plus/Minus and Defensive Plus/Minus
- Win shares: Contribution of each player into wins of the team.
- VORP – Value Over Replacement Level

Where to find the ratings?

Plus/Minus

- Best lineup by plus/minus
  https://stats.nba.com/lineups/traditional/
- ESPN: Offensive Real Plus/Minus, Defensive real Plus/Minus, RPM and Wins share (from 2014) https://www.espn.com/nba/statistics/rpm/_/sort/RPM
- Plus minus by player
  https://www.basketball-reference.com/players/d/duranke01/gamelog/2012/

Other Metrics

- WINVAL by Wayne Winston: The exact method of calculating tha Adjusted P/M is not public, however.

Margin of victory = sum(Rating of all 5 home team players) - sum(Rating of all 5 away team players)